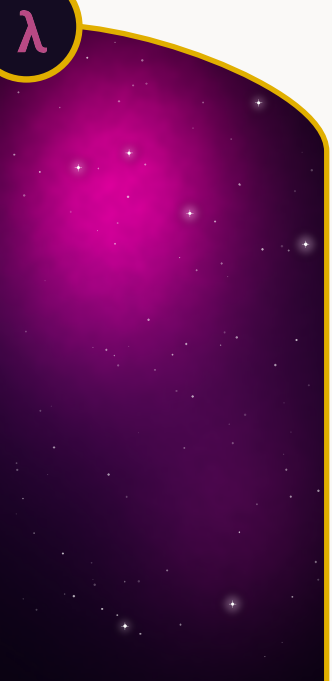


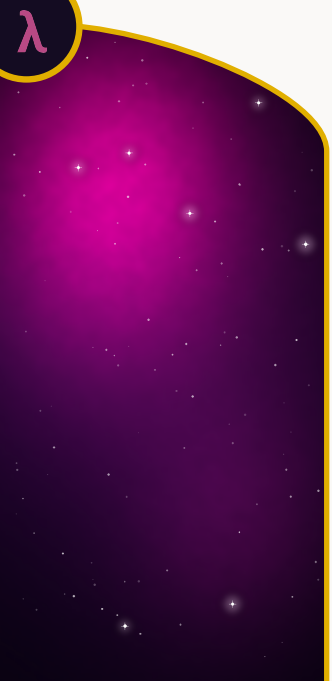
L2B: Orders of Growth

CS1101S: Programming Methodology

Boyd Anderson

21st August, 2026

Version 1.1 — Built: 2026-08-21 06:11:58



Outline

Announcements

Tree Recursion (Textbook 1.2.2)

Orders of Growth (Textbook 1.2.3)

Calendars & Absence Policy

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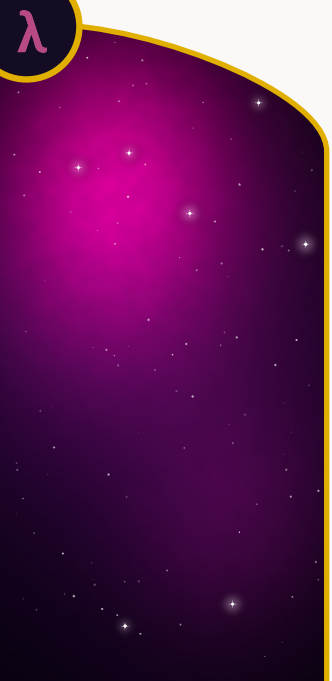
Keep your MCs — only submit them at the end of semester if you exceed the allowance above.

Studio Attendance and Participation

- Attendance and Participation (5% of CS1101S)
  - Studios are small communities of learners; your presence in the sessions is needed to be an effective member
  - 3% for Studio attendance
  - 2% for Studio participation; Avengers assess your participation, based on effort
  - The ideal Studio member prepares well for the meetings, contributes by answering any questions thoughtfully, asks good questions, and tries to help classmates in mastering the material
- Studio Performance XP
  - Maximal 500 XP per Studio session
  - Here is where you can shine, by helping others, contributing to discussions, going the extra mile



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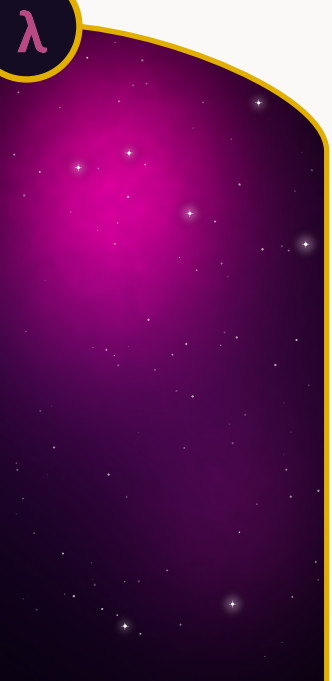

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Evaluating fib(5)

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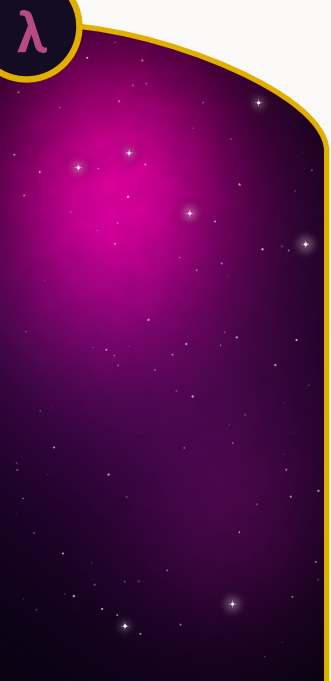

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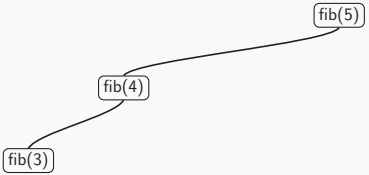

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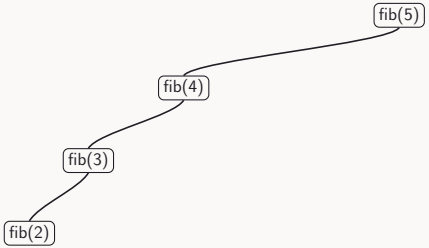

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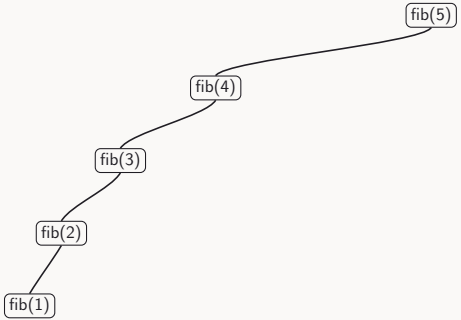

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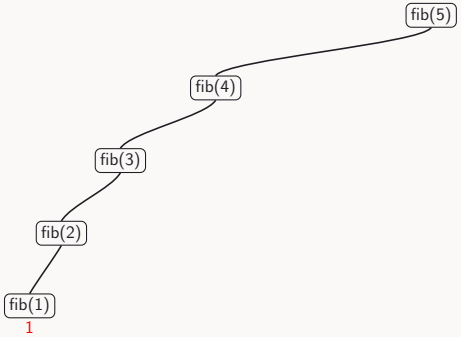

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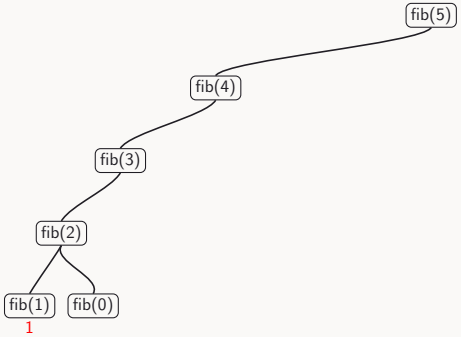

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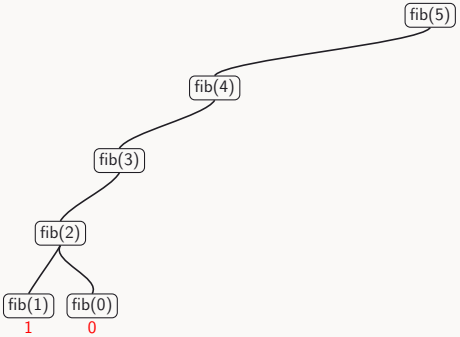

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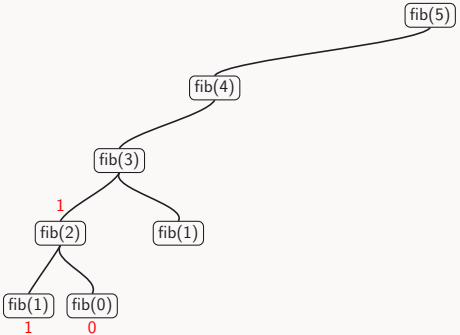

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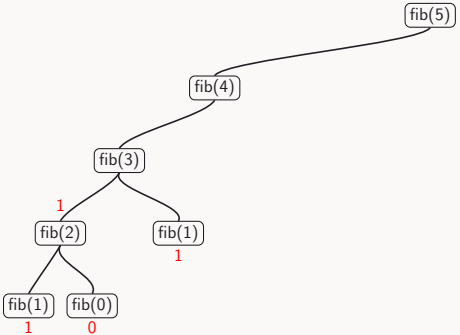

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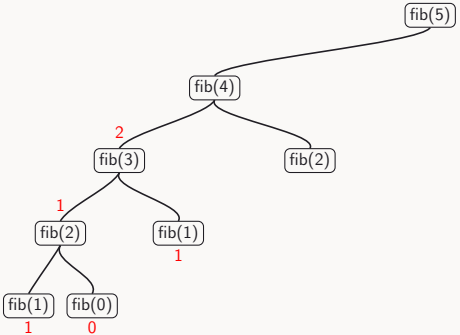

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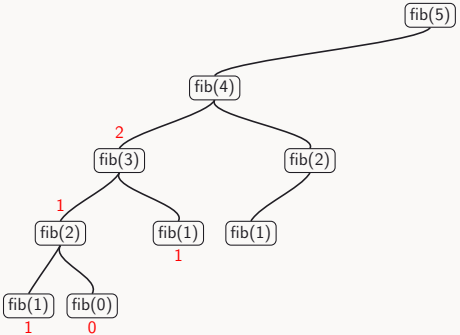

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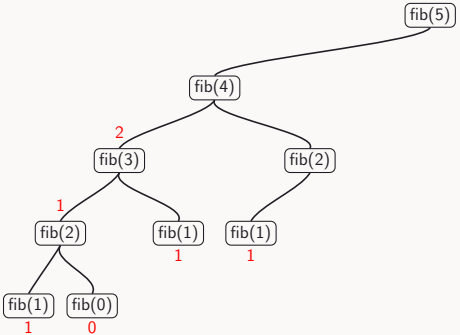

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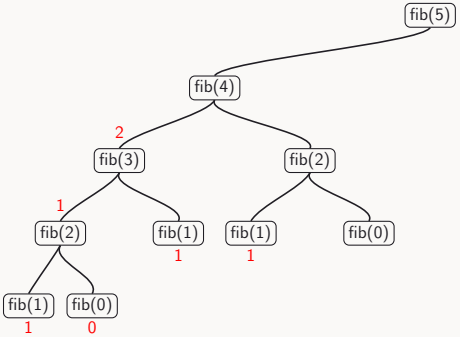

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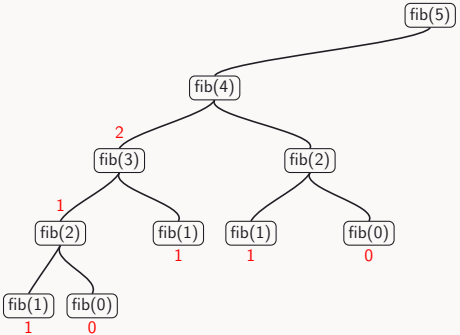

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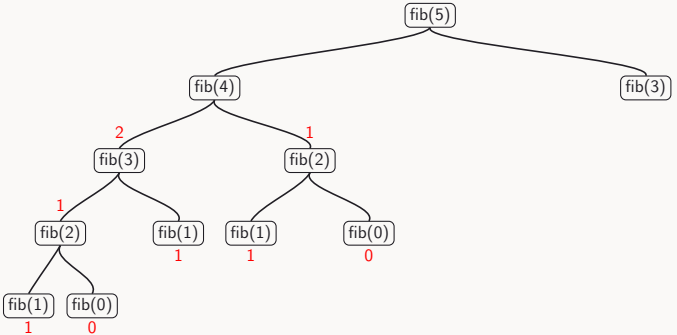

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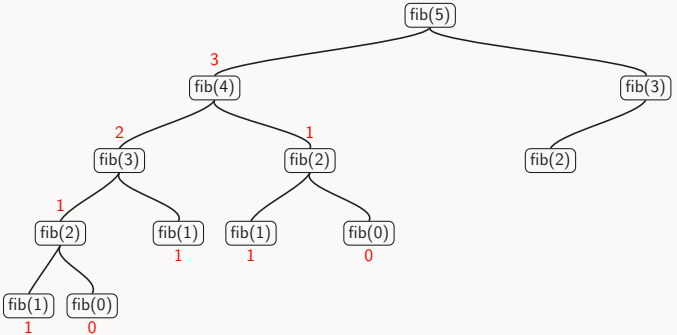

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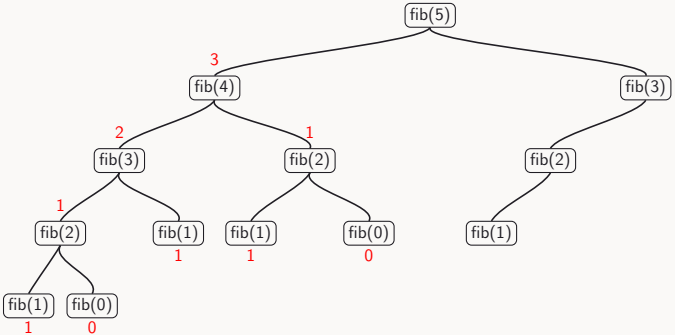

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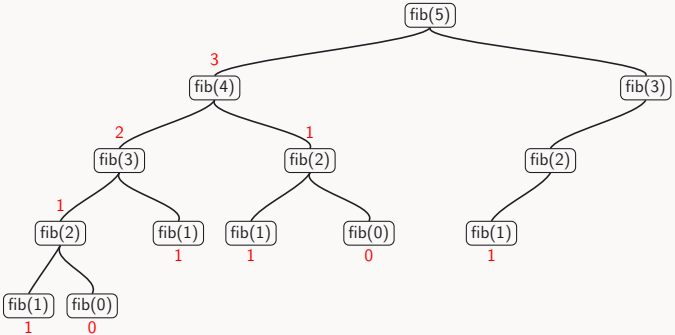

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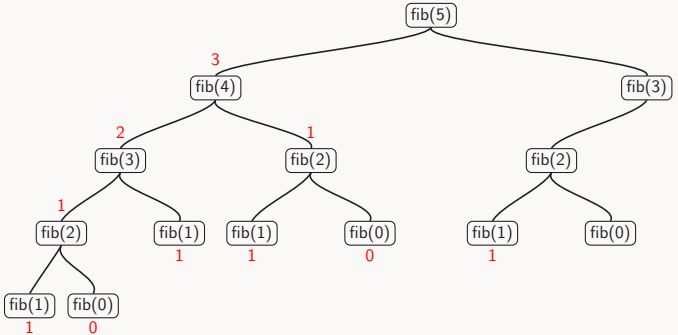

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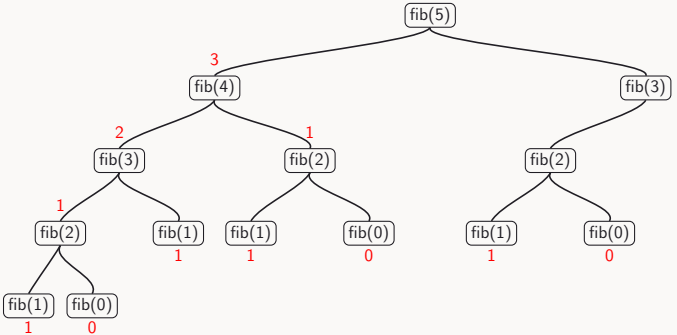

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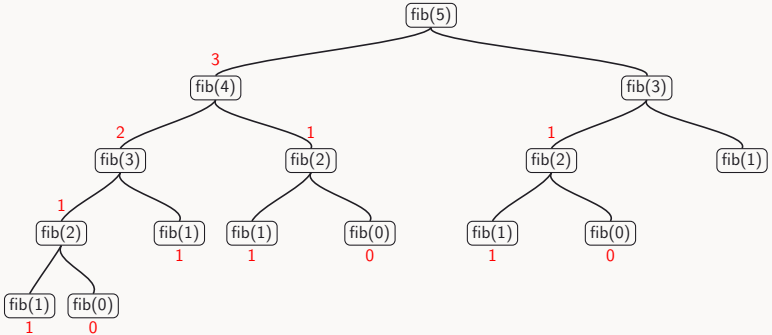

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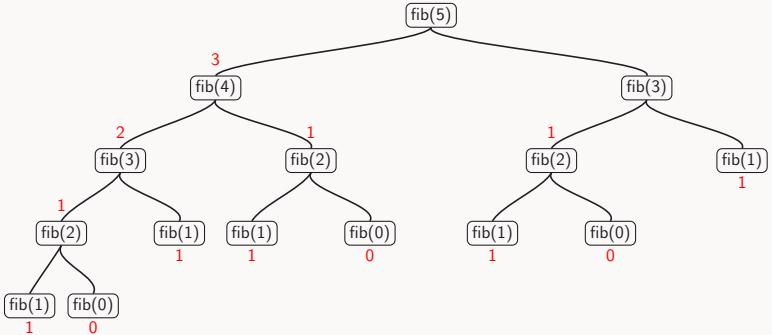

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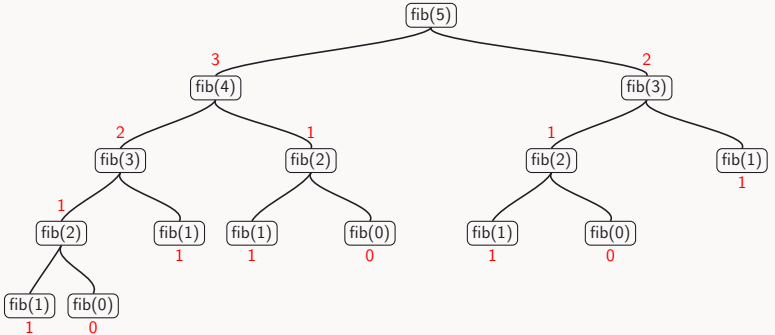

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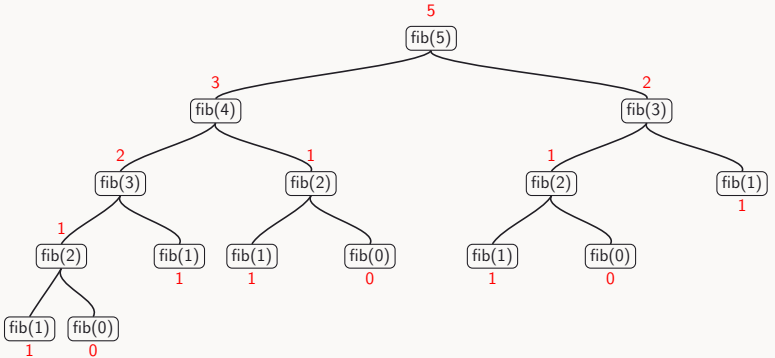

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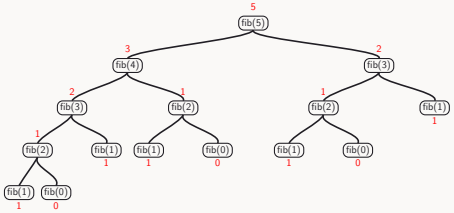

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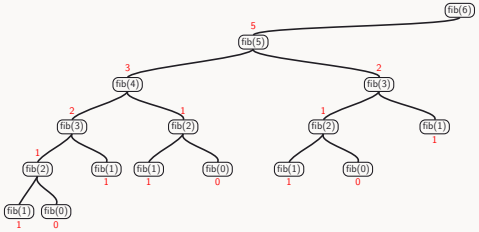

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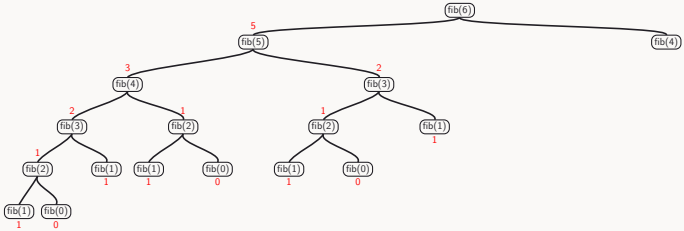

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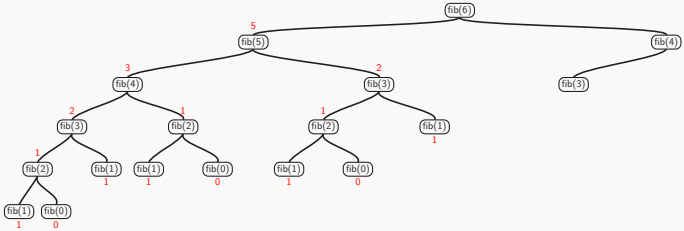

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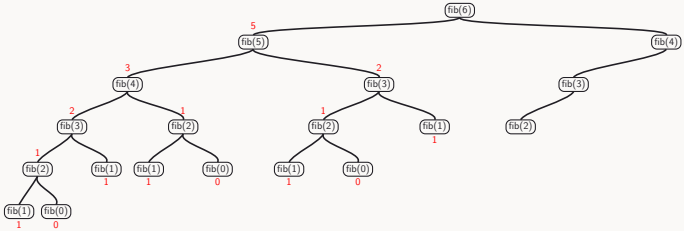

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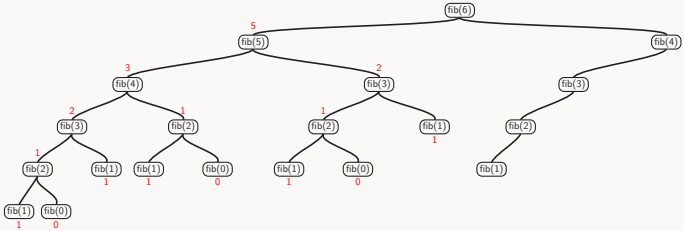

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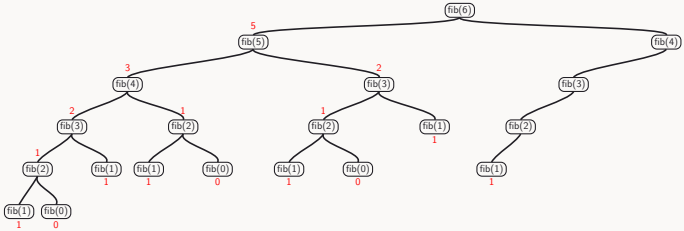

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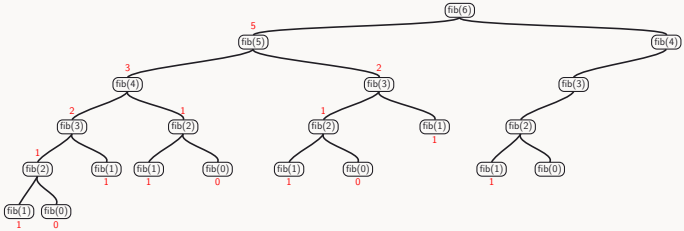

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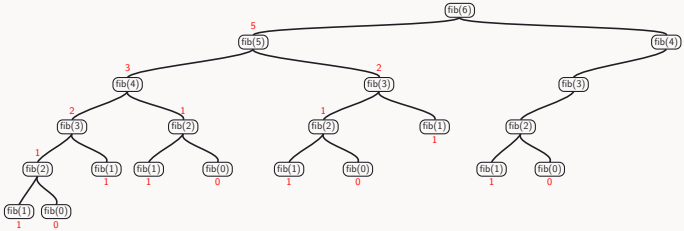

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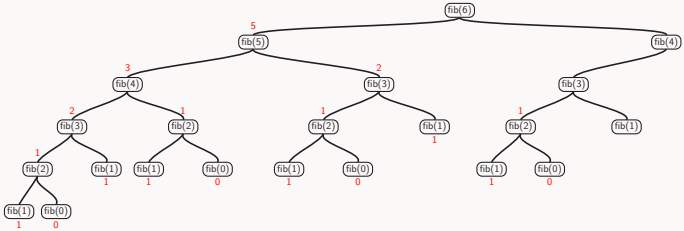

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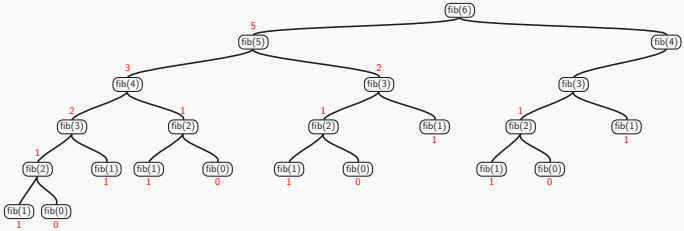

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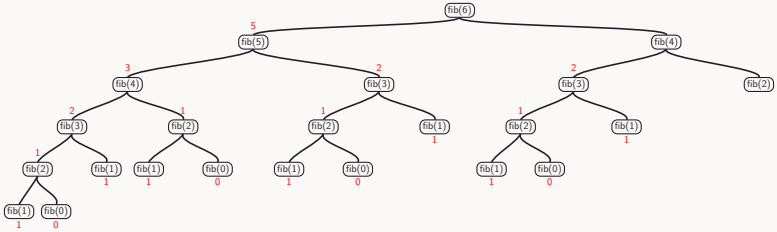

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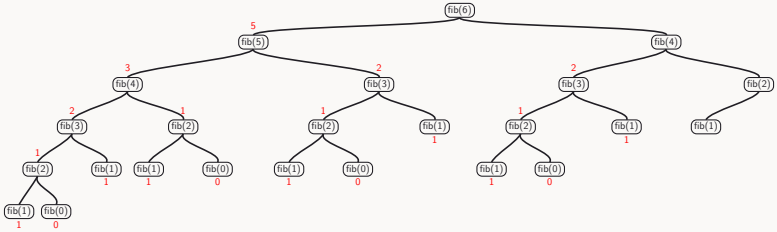

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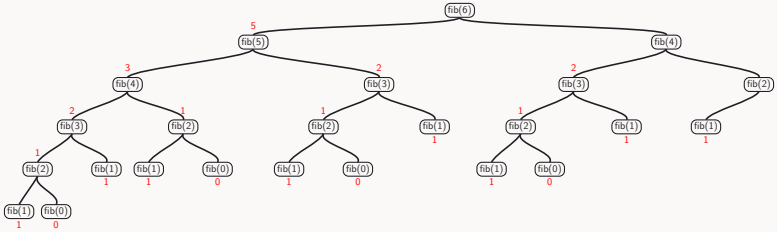

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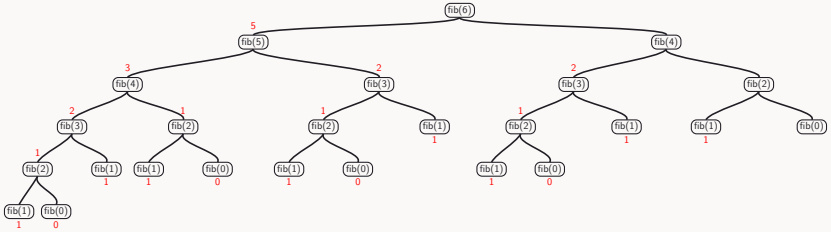

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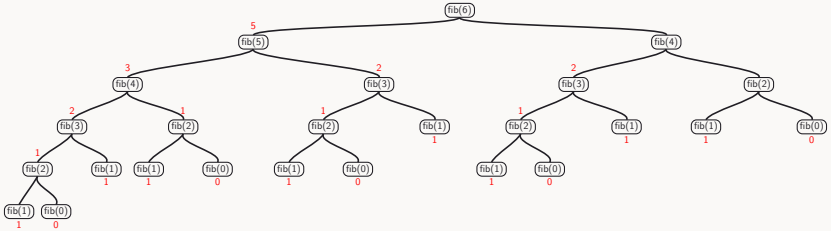

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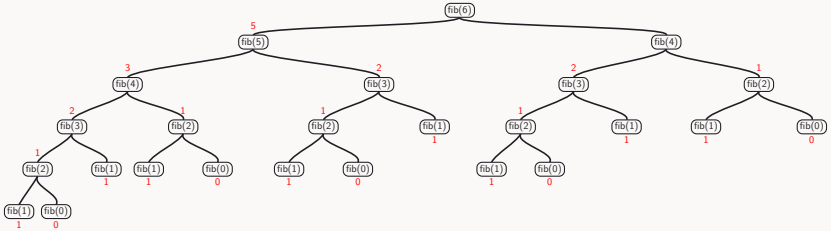

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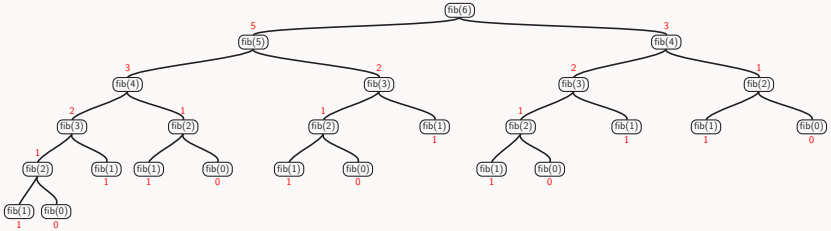

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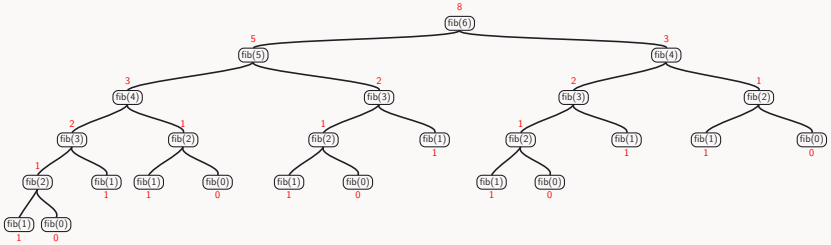

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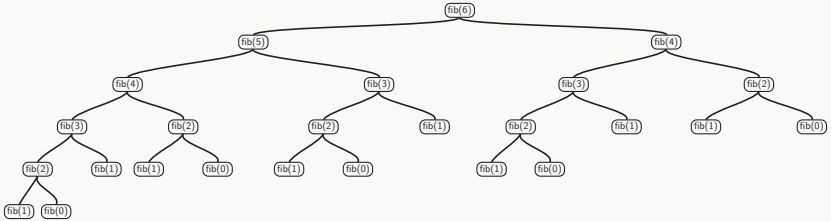

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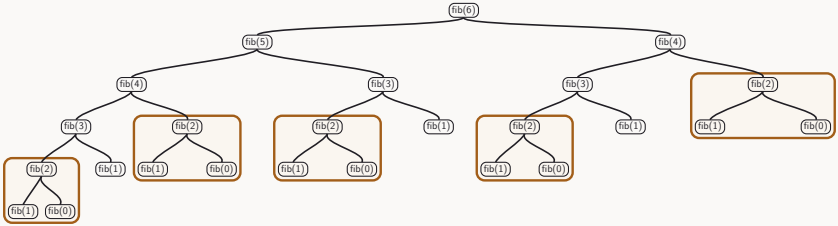

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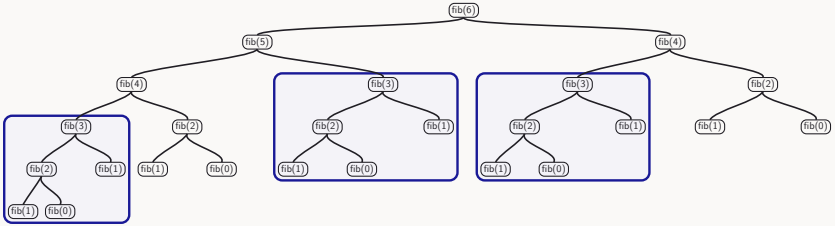

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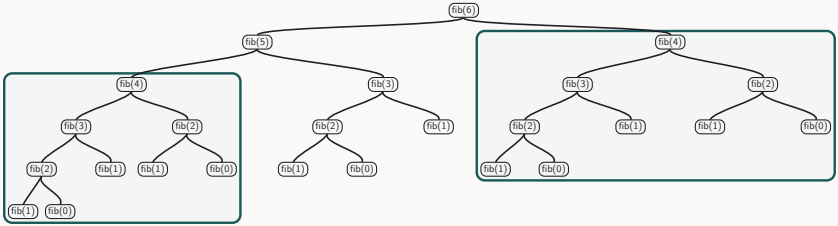
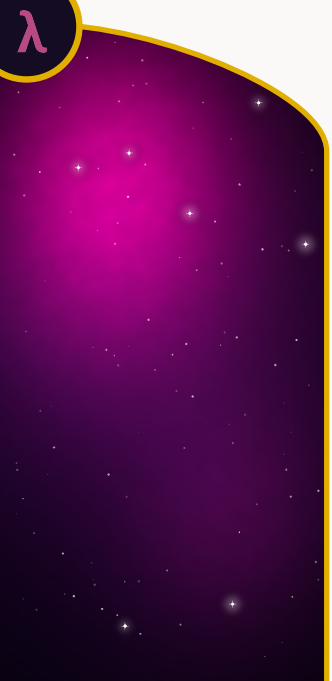


Example: Computing F(n) using fib(n)

- Number of leaves in the recursion tree for fib(n) is F(n + 1)
  - fib(10) needs to visit $F(11) = 89$ leaves
  - fib(20) needs to visit $F(21) = 10946$ leaves
  - fib(40) needs to visit $F(41) = 165580141$ leaves
  - fib(100) needs to visit $F(101) = 573147844013817084101$ leaves

Time for Evaluating fib(n)

- Time for exploring the tree grows with size of tree



Time for Evaluating fib(n)

- Time for exploring the tree grows with size of tree
- Tree for fib(n) has F(n + 1) leaves, where
  - $F(n) = \left\lfloor \frac{\varphi^n}{\sqrt{5}} + \frac{1}{2} \right\rfloor$, and $\varphi = \frac{1+\sqrt{5}}{2}$

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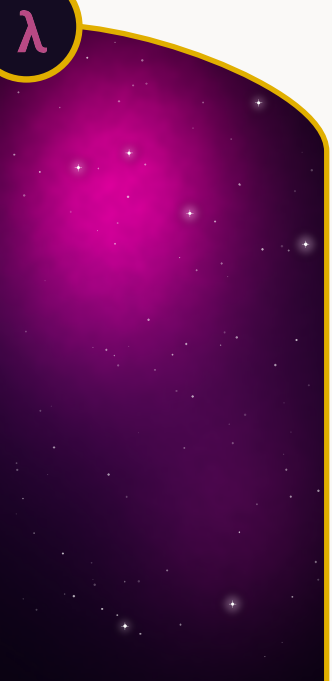
Can we write an efficient iterative function that computes F(n)?



Fibonacci Numbers (iterative version)

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def fib(n):
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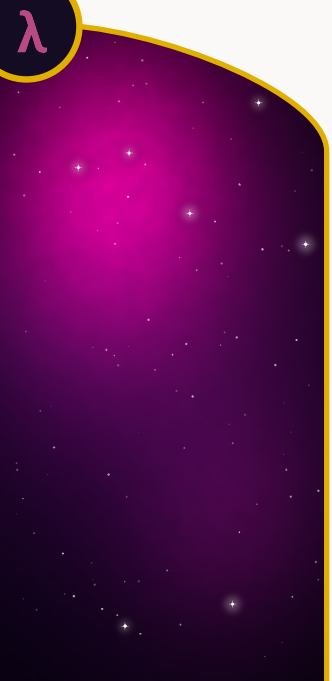
- How efficient is this program?

Wait. What does it even mean to be efficient?



Outline

Announcements

Tree Recursion (Textbook 1.2.2)

Orders of Growth (Textbook 1.2.3)

A Tale of Two Programmers

Prof Martin
writes beautiful, elegant code

Prof Sanka
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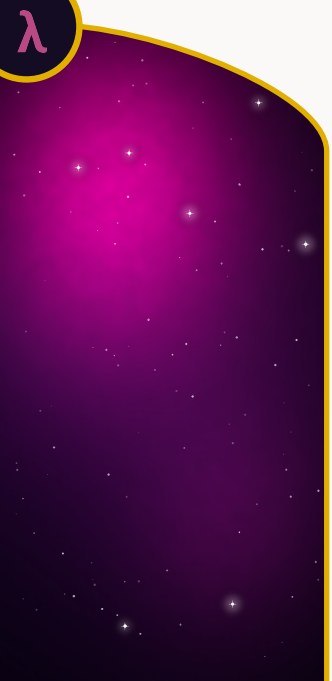
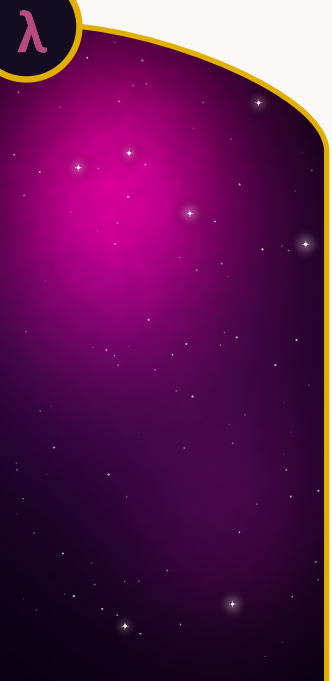
Both are asked to solve the *same* problem.



How Do We Compare Their Programs?

- Readability



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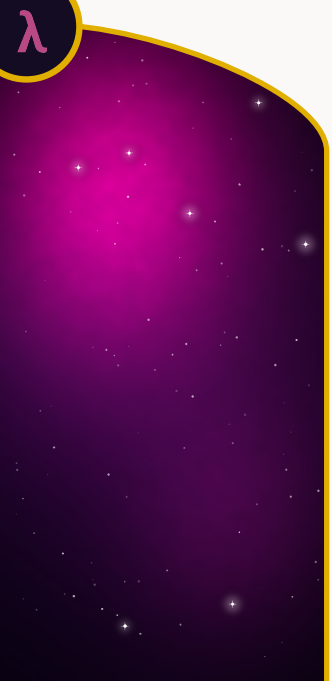

How Do We Compare Their Programs?

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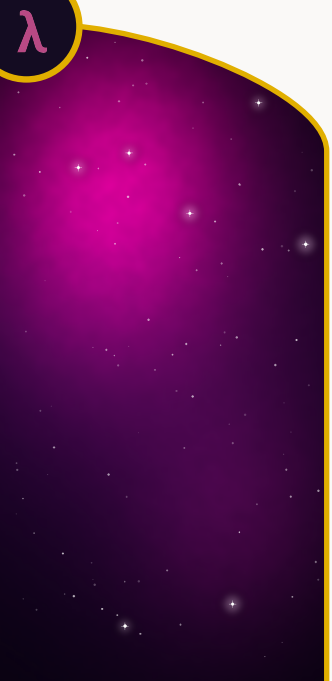
- Style
- Elegance



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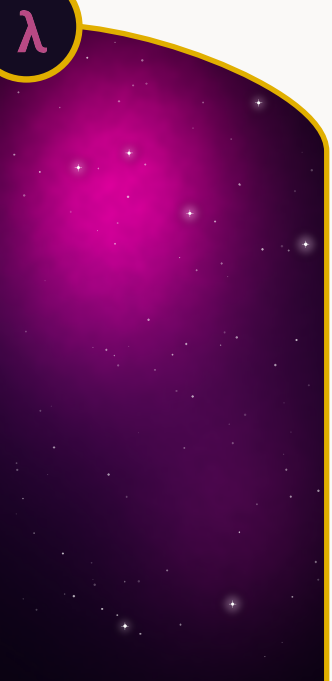

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Wait, how do we compare programs?

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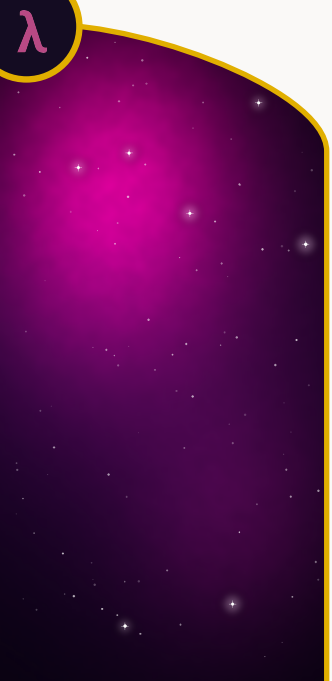
Wait, how do we compare programs?

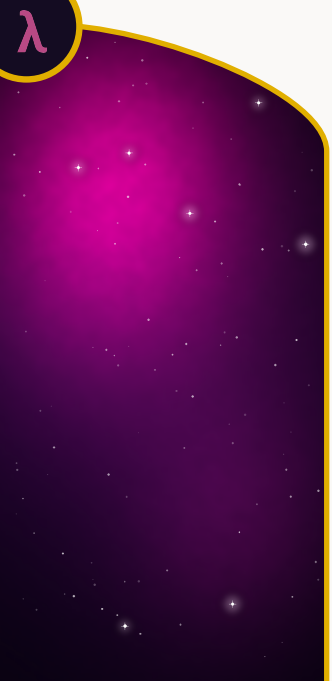
Today: two measurable dimensions

Time: how long does the program run
Space: how much memory does it need



Recall the Factorial Function

- Recursive definition
  - $n! = 1$ if $n = 1$
  - $= n(n-1)!$ if $n > 1$



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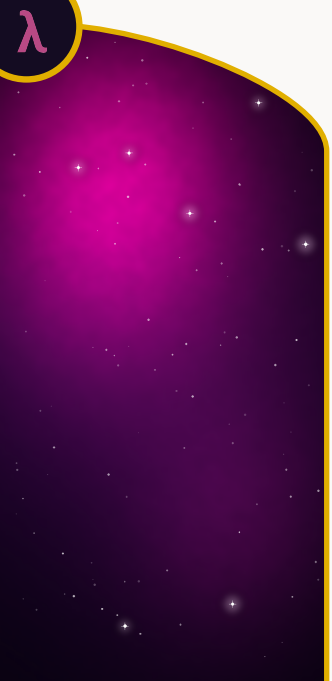
```

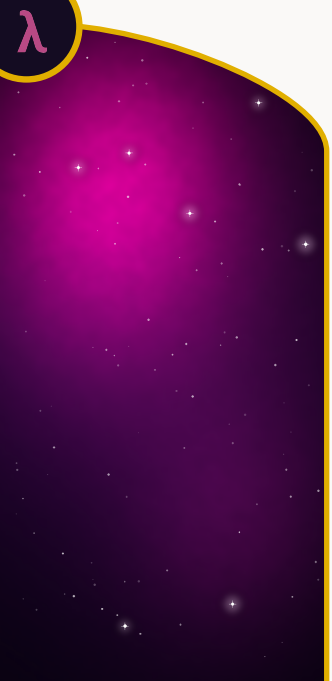
- Example call:

```
print(factorial(4))
```



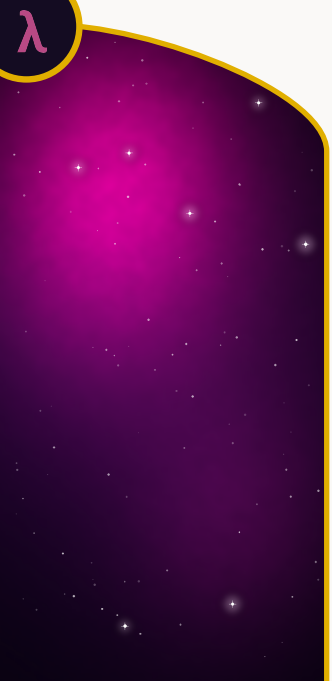
Time for Calculating factorial(n)

```
factorial(4)
```



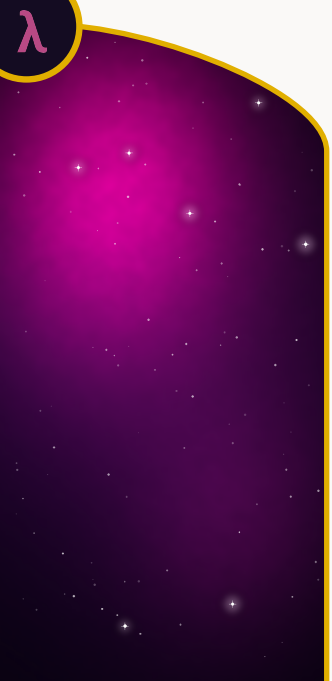
Time for Calculating factorial(n)

```
factorial(4)
4 * factorial(3)
```



Time for Calculating factorial(n)

```
factorial(4)
4 * factorial(3)
4 * (3 * factorial(2))
```



Time for Calculating factorial(n)

```
factorial(4)
4 * factorial(3)
4 * (3 * factorial(2))
4 * (3 * (2 * factorial(1)))
```

Time for Calculating factorial(n)

```
factorial(4)
4 * factorial(3)
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```

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4 * (3 * (2 * 1))
4 * (3 * 2)
```

Time for Calculating factorial(n)

```
factorial(4)
4 * factorial(3)
4 * (3 * factorial(2))
4 * (3 * (2 * factorial(1)))
4 * (3 * (2 * 1))
4 * (3 * 2)
4 * 6
```

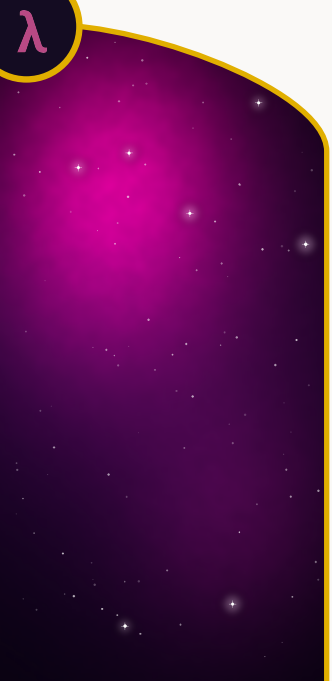
Time for Calculating factorial(n)

```
factorial(4)
4 * factorial(3)
4 * (3 * factorial(2))
4 * (3 * (2 * factorial(1)))
4 * (3 * (2 * 1))
4 * (3 * 2)
4 * 6
24
```

Time for Calculating factorial(n): Observation

Observation

Number of operations grows linearly proportional to n



Space for Calculating factorial(n)

```
factorial(4)
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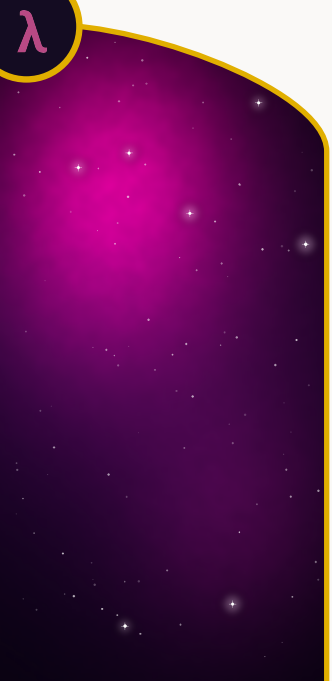

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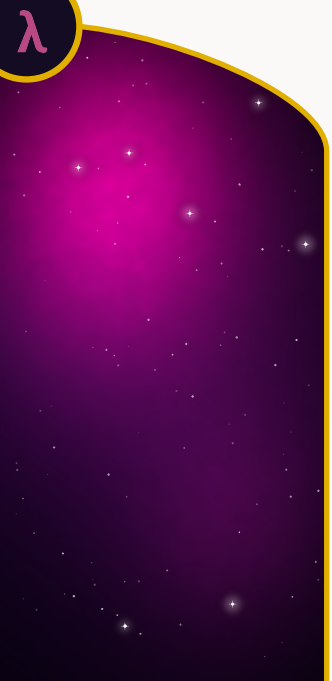

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factorial(4)
4 * factorial(3)
4 * (3 * factorial(2))
```

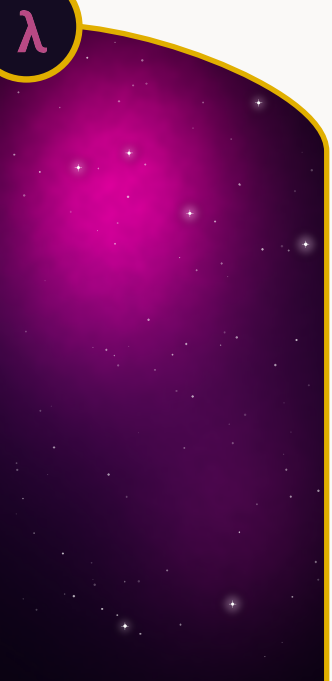

Space for Calculating factorial(n)

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factorial(4)
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4 * 6
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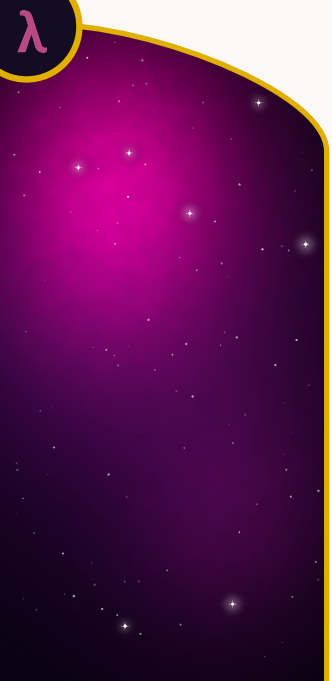
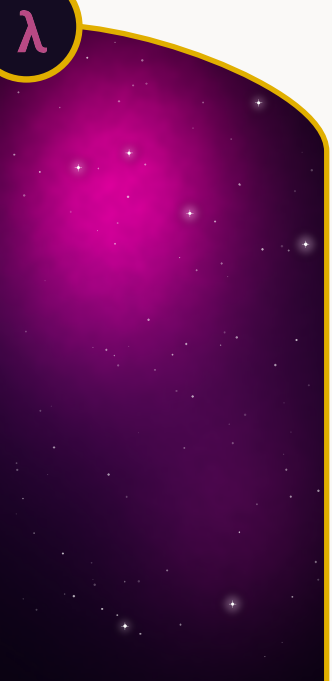


Doubling the Argument of factorial (n = 2)

```
factorial(2)
```



Doubling the Argument of factorial (n = 2)

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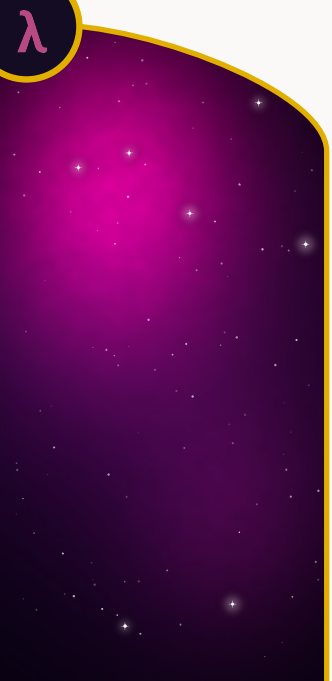
factorial(2)
2 * factorial(1)
```



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factorial(2)
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2 * 1
```

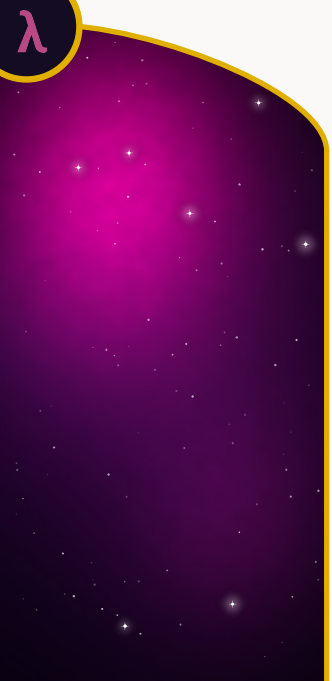

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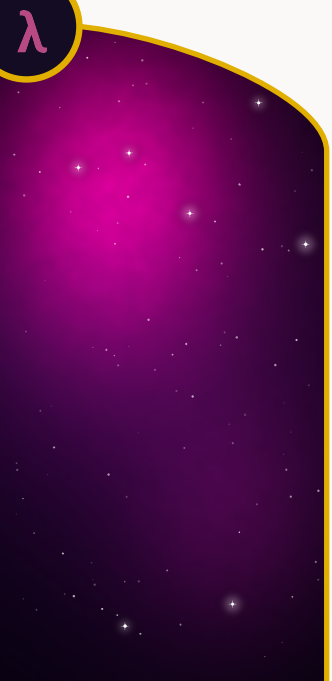
2
```



Doubling the Argument of factorial (n = 4)

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factorial(4)
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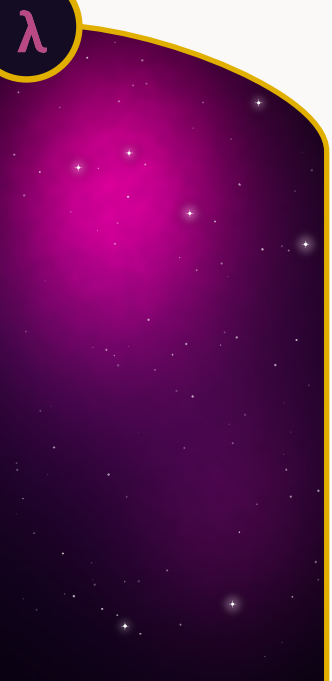
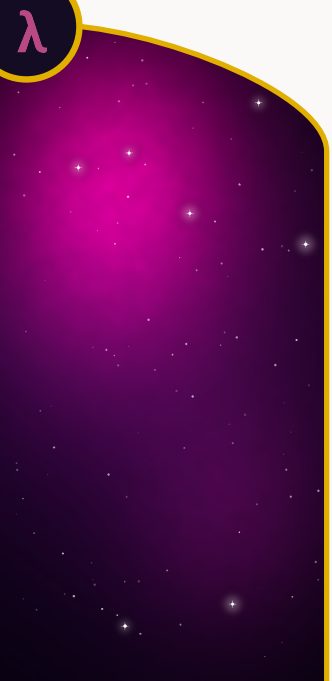


Doubling the Argument of factorial (n = 4)

```
factorial(4)
4 * factorial(3)
```



Doubling the Argument of factorial (n = 4)

```
factorial(4)
4 * factorial(3)
4 * (3 * factorial(2))
```

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factorial(4)
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4 * 6
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4 * (3 * (2 * 1))
4 * (3 * 2)
4 * 6
24
```

Doubling the Argument of factorial: Observation

Time

The number of steps “roughly” doubles. The factorial function runs in a time (linearly) proportional to the argument n.

Doubling the Argument of factorial: Observation

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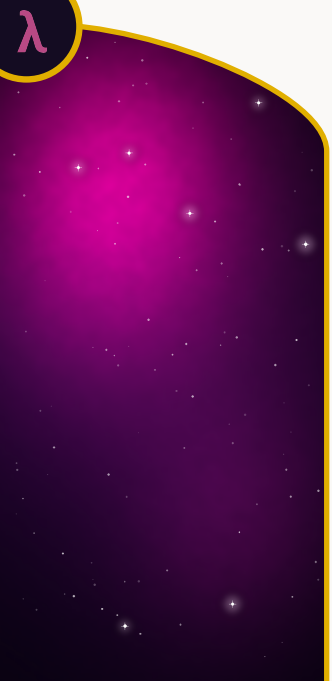
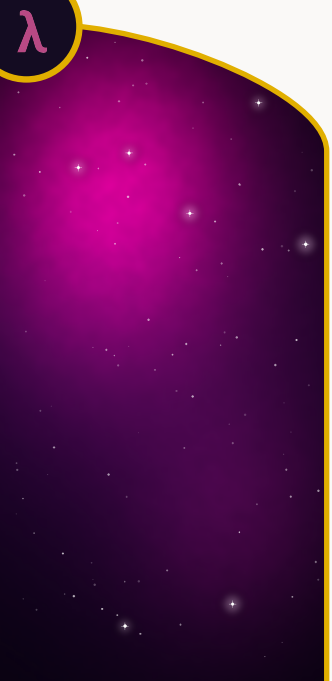
Space

The number of deferred operations also “roughly” doubles. It has a space requirement (linearly) proportional to the argument n.



Orders of Growth

- Linear growth
  - The factorial function runs in a time (linearly) proportional to the argument n



Orders of Growth

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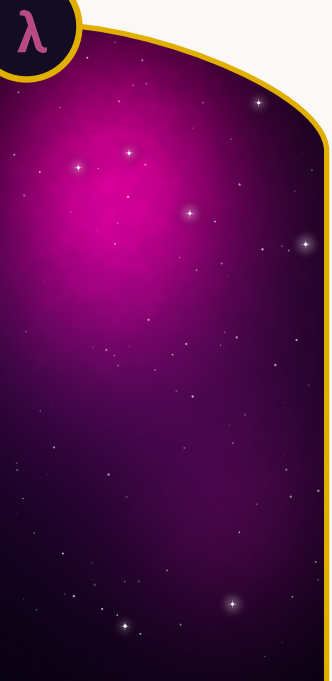
- Exponential growth
  - The fib function runs in a time that grows exponentially with the argument n



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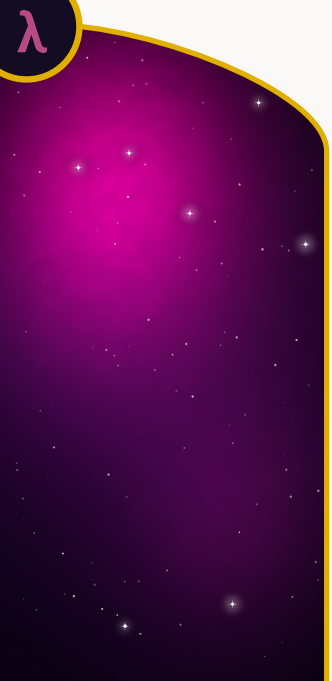
What exactly do we mean by the above?

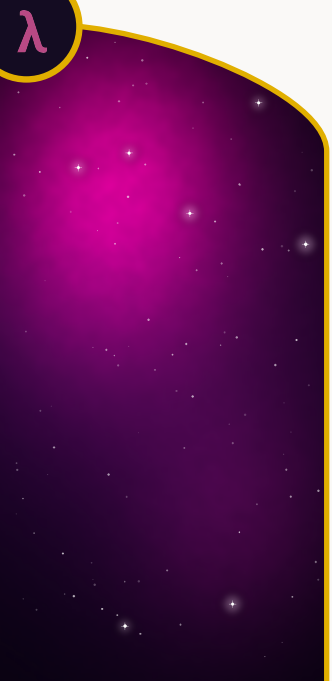
Purpose

- Rough measure
  - We are interested in a rough measure of resources used by a computational process, with respect to the problem size
- Abstraction
  - "Order of growth" is an abstraction technique
  - We decide to ignore details that we deem irrelevant
  - Examples: the processor speed of the computer, the programming environment, the programming language, or minor differences in programming style



The Resource Function

- For a program P solving a problem of size n, let $r(n)$ denote the amount of a *resource* (time or space) that P needs



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The Resource Function

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- Recall Prof Martin and Prof Sanka: both write a program for the *same* problem
- Plug in Martin's program: $r(n)$ tells us its resource use
- Plug in Sanka's program instead: $r(n)$ now tells us Sanka's program's resource use, same symbol, different program

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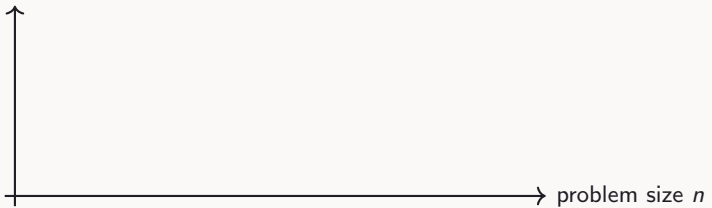

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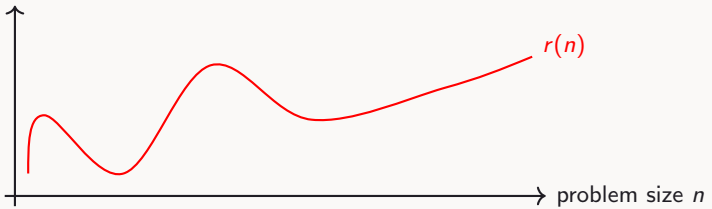

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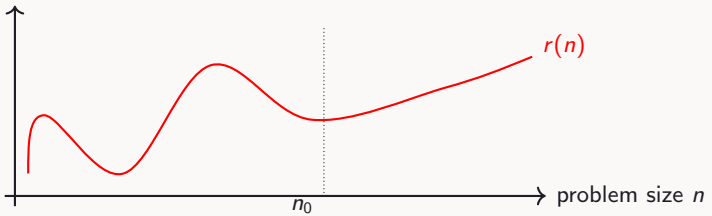

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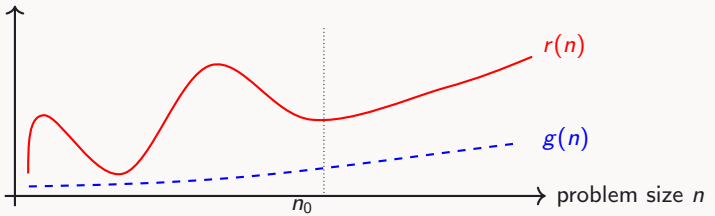

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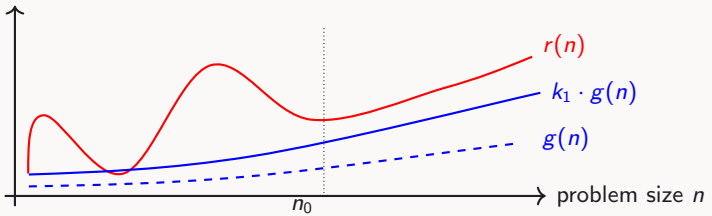

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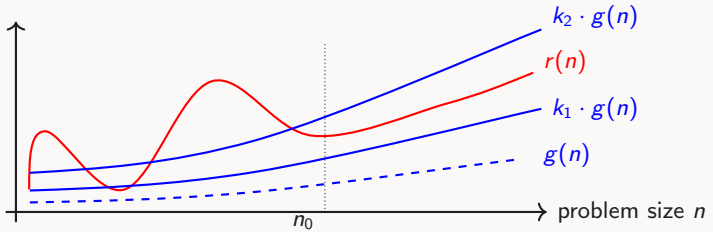

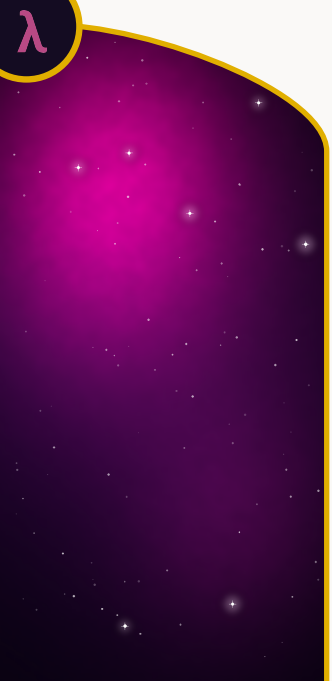

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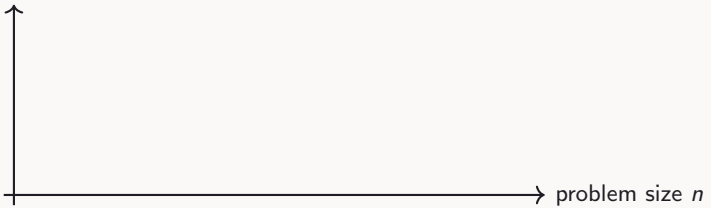

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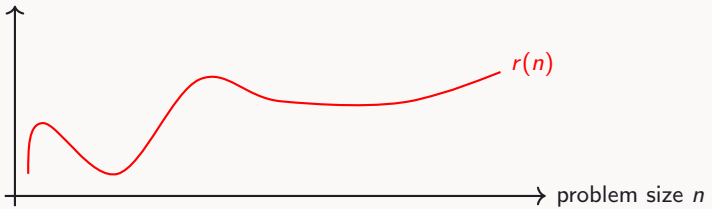

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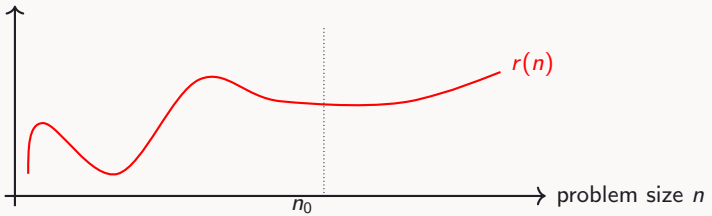

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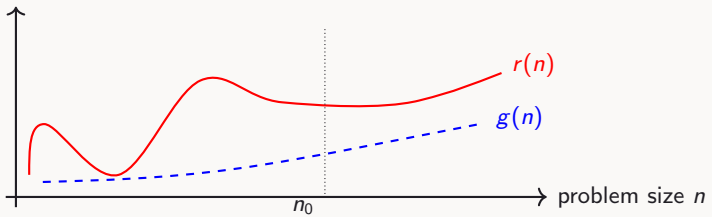

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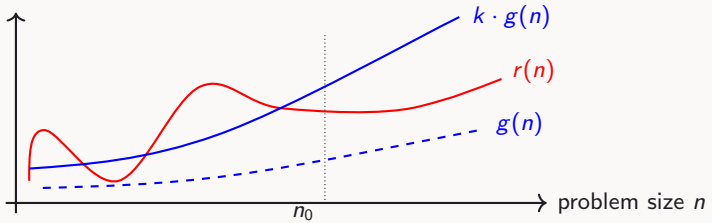

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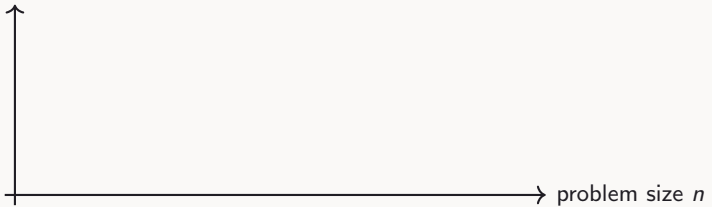

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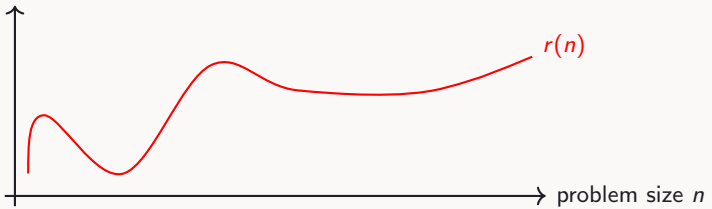

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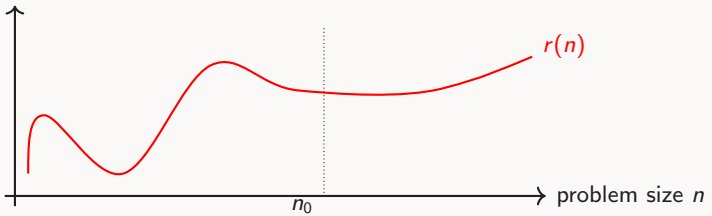

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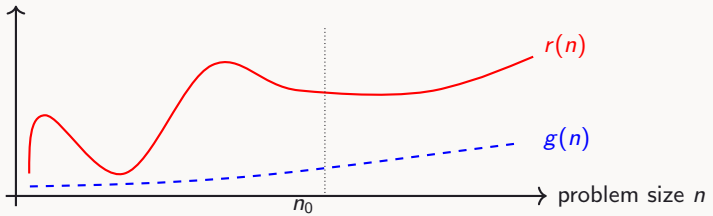

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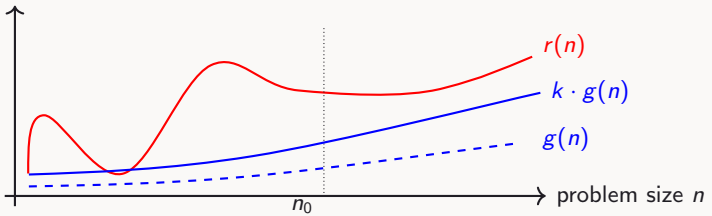
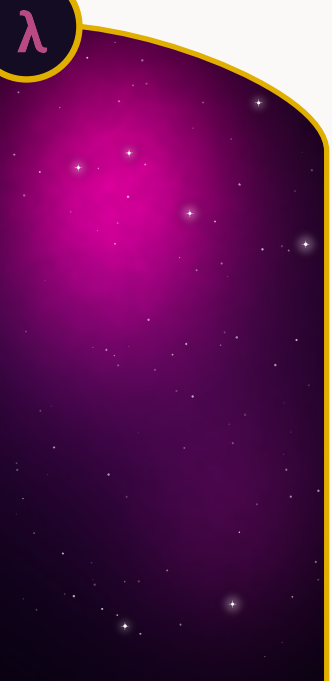
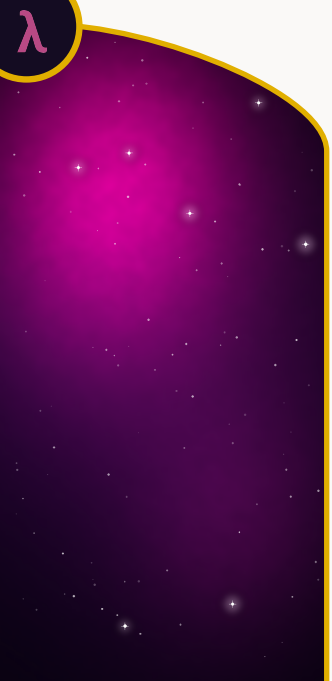




Do Constants Matter?

- Let's say r has order of growth $\Theta(n^2)$



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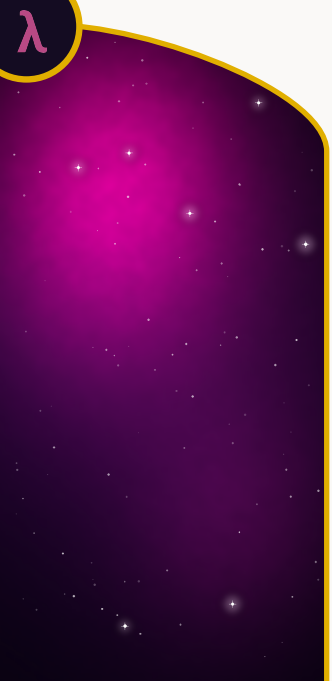
- Does r also have order of growth $\Theta(0.5n^2)$?



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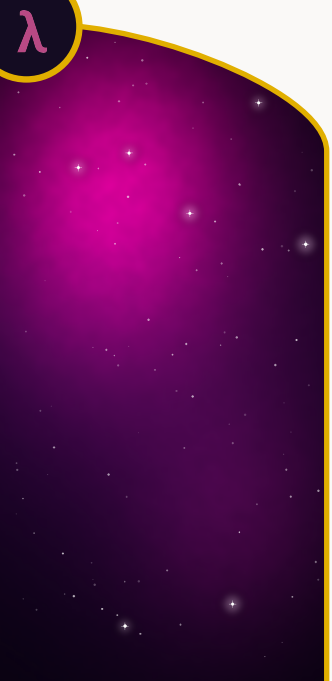
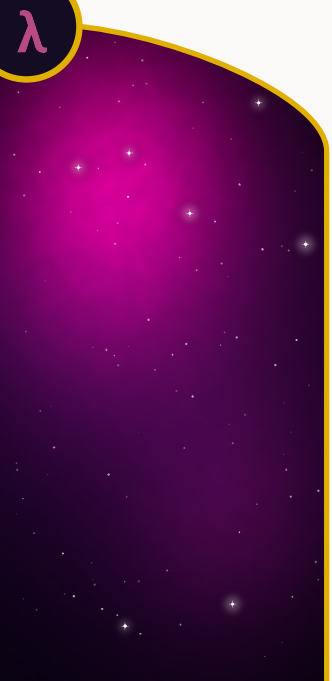
- Constants don't matter
  - Because we can freely choose k, k_1, k_2



Do Minor Terms Matter?

- Let's say r has order of growth $O(n^2)$



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- Does r also have order of growth $O(n^2 + 40n - 5)$?
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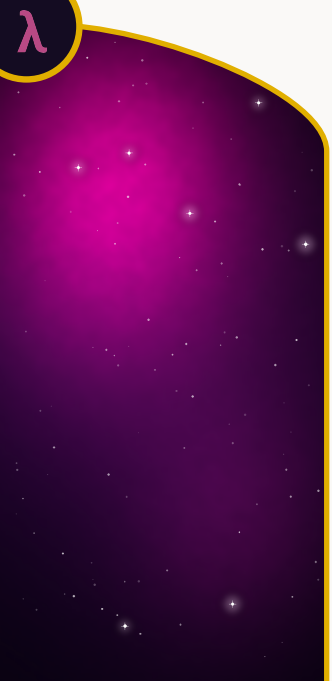
  - Because we can adjust n_0, k, k_1, k_2 such that the minor terms are overruled



Some Common g(n)

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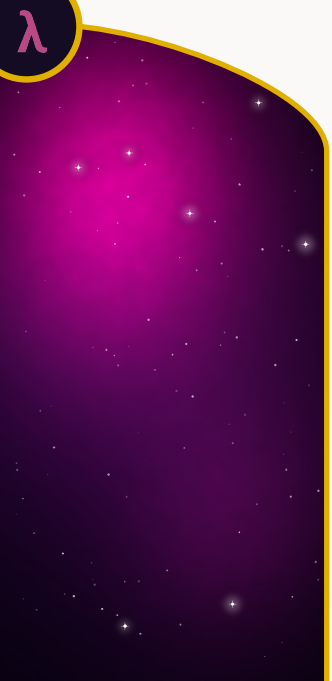

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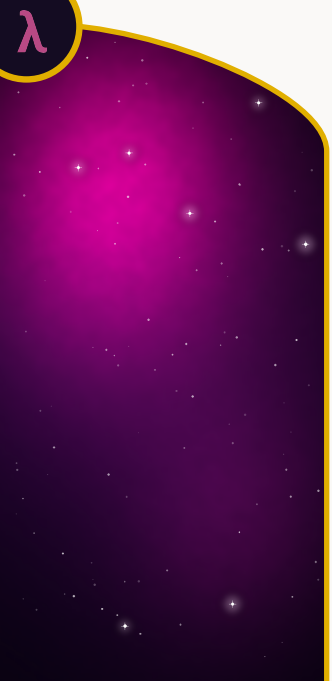

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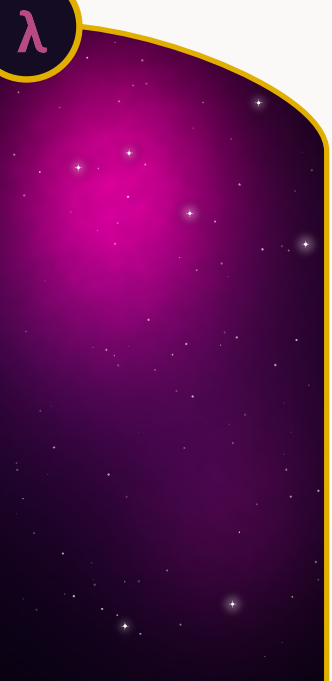

Some Common g(n)

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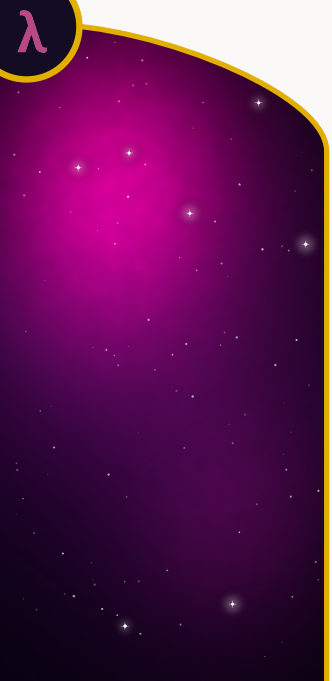
- n
- n log n
- n^2



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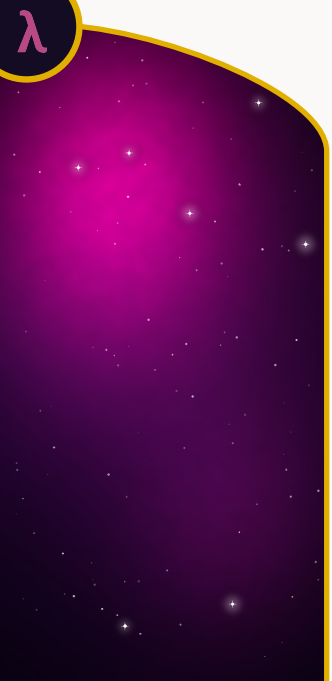

Some Common g(n)

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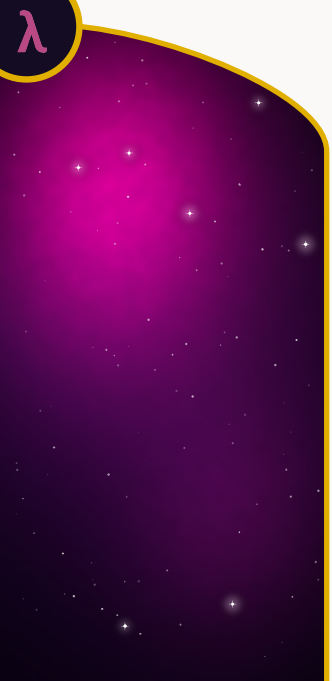
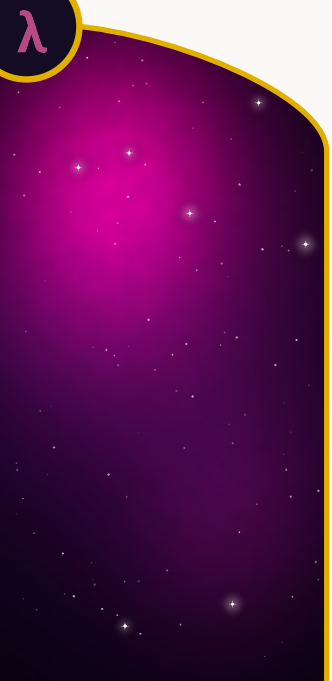
- 2^n



Growing Pains



Growing Pains

It is now time (or space) to go to the Source Academy!

Apologies, I am running out of jokes here already and we are only in week 2!

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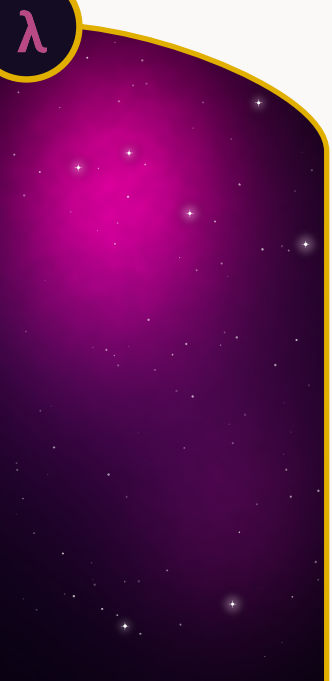
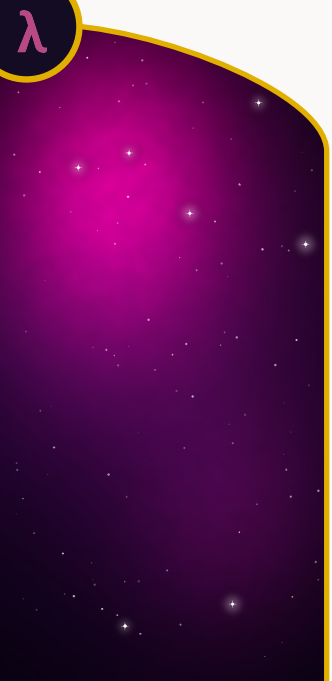
(No need to finalize your Check-In submission. Please stop working as soon as we call time!)



Example Orders of Growth

- Linear time
  - The `factorial` function runs in a time that has order of growth $\Theta(n)$, where n is the function argument



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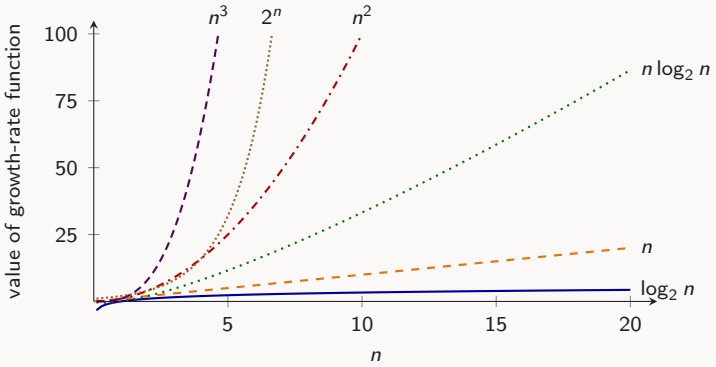
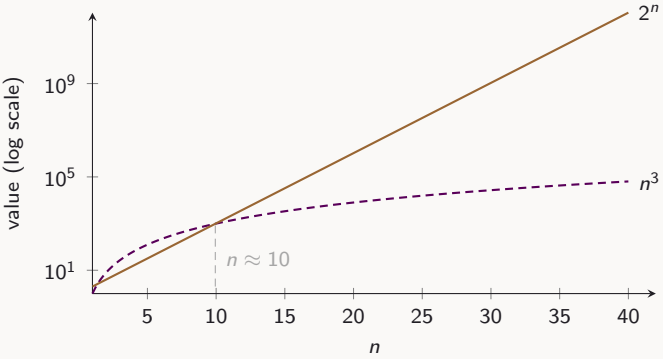
Comparing Orders of Growth: Table

	n					
Function	10	100	1,000	10,000	100,000	1,000,000
1	1	1	1	1	1	1
$\log_2 n$	3	6	9	13	16	19
n	10	10^2	10^3	10^4	10^5	10^6
$n \log_2 n$	30	664	9,965	10^5	10^6	10^7
n^2	10^2	10^4	10^6	10^8	10^{10}	10^{12}
n^3	10^3	10^6	10^9	10^{12}	10^{15}	10^{18}
2^n	10^3	10^{30}	10^{301}	$10^{3,010}$	$10^{30,103}$	$10^{301,030}$

Comparing Orders of Growth: Graph



Comparing Orders of Growth: n^3 vs 2^n



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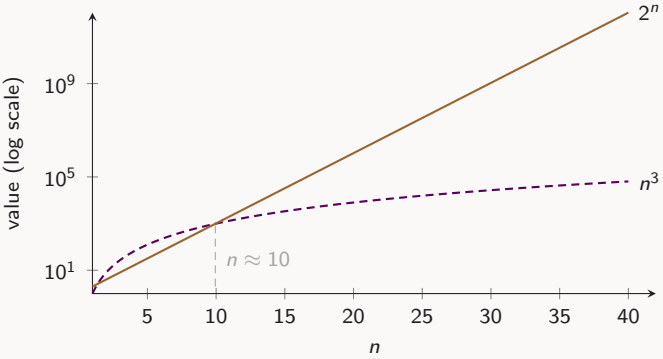
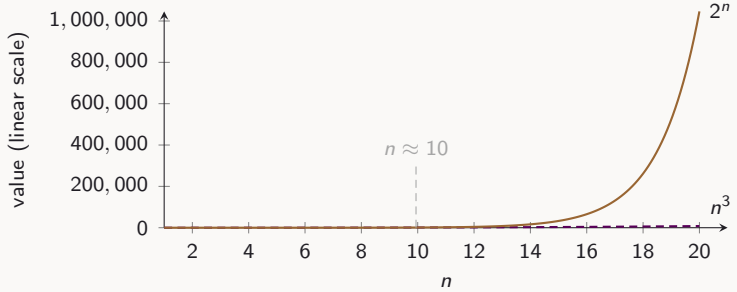


They cross around $n \approx 10$ and past that, 2^n pulls away from n^3 ever faster, and the gap grows without bound.

n^3 vs 2^n: Linear Scale



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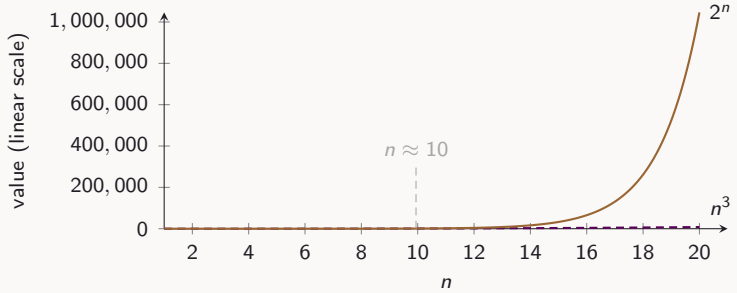

On a linear scale, n^3 all but disappears next to 2^n once n grows a little past the crossover this is exactly why we use order of growth instead of trying to compare raw numbers.

How to Calculate “Big Oh/Theta/Omega”

- Topic of Algorithm Analysis (CS3230)
- For us:
  - Identify the basic computational steps
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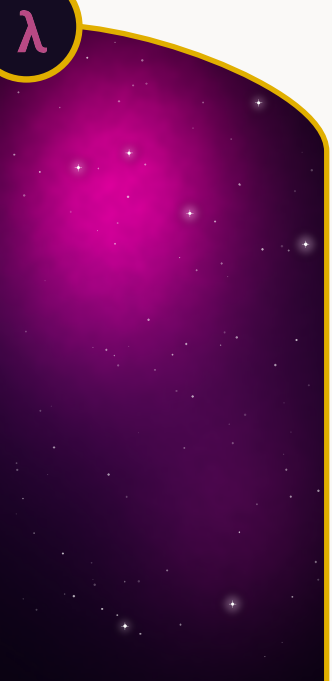
Why tightest?

It's true that Usain Bolt can run 100m in under 10 *minutes*... but that's not meaningful. What we actually care about is that he can do it in under 10 *seconds*.



Summary

- Resources for computational processes: time and space
- Big Theta, Big Oh, Big Omega
  - Provide abstractions or “rough” measures of resources used with respect to problem size